

## Exercise 6: Library Management System

### CODE:

```
class Book {
    int bookId;
    String title;
    String author;
    Book(int bookId, String title, String author) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
    }
    void display() {
        System.out.println(bookId + "\t" + title + "\t\t" + author);
    }
}

public class Main {
    // Linear Search
    static Book linearSearch(Book[] books, String title) {
        for (Book book : books) {
            if (book.title.equalsIgnoreCase(title)) {
                return book;
            }
        }
        return null;
    }
    // Binary Search
    static Book binarySearch(Book[] books, String title) {
        int low = 0;
        int high = books.length - 1;
```

```

while (low <= high) {
    int mid = (low + high) / 2;
    int result = books[mid].title.compareToIgnoreCase(title);
    if (result == 0) {
        return books[mid];
    } else if (result < 0) {
        low = mid + 1;
    } else {
        high = mid - 1;
    }
}
return null;
}

public static void main(String[] args) {
    // Unsorted array for Linear Search
    Book[] books = {
        new Book(101, "Java", "James Gosling"),
        new Book(102, "Python", "Guido van Rossum"),
        new Book(103, "C", "Dennis Ritchie"),
        new Book(104, "Data Structures", "Mark Allen")
    };

    System.out.println("Linear Search Result");
    Book result1 = linearSearch(books, "Python");
    if (result1 != null) {
        System.out.println("ID\tTitle\t\tAuthor");
        result1.display();
    }

    // Sorted array for Binary Search (sorted by title)
    Book[] sortedBooks = {

```

```
        new Book(103, "C", "Dennis Ritchie"),
        new Book(104, "Data Structures", "Mark Allen"),
        new Book(101, "Java", "James Gosling"),
        new Book(102, "Python", "Guido van Rossum")
    };

    System.out.println("\nBinary Search Result");

    Book result2 = binarySearch(sortedBooks, "Python");
    if (result2 != null) {
        System.out.println("ID\tTitle\t\tAuthor");
        result2.display();
    }
}
```

**OUTPUT:**

```
Linear Search Result
ID  Title      Author
102 Python      Guido van Rossum

Binary Search Result
ID  Title      Author
102 Python      Guido van Rossum
```